SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Scenario-Based Questions

Course Code:	DSA0405	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO4 – Scenario-Based Questions	Assessment:	Assessment Tool 1– Scenario-Based Questions
Weightage:	40% of CIA	Total Marks:	100
CO4 Statement:	Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)		
Student Name:	JOTHIKA M	Reg. No.:	192324235
Section / Batch:	2023-B.Tech(AI & DS)	Date:	17-08-26

Code of Conduct

I Jothika M (192324235) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Jothika M

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	**Thorough understanding; clearly identifies all key issues**	**Good understanding with most issues identified**	**Basic understanding; some issues missed**	**Poor understanding; problem not identified correctly**	
Application of Concepts	25	**Effectively applies relevant concepts to analyze the case deeply**	**Applies appropriate concepts with minor gaps**	**Limited application of concepts**	**Fails to apply relevant concepts**	
Analysis	20	**In-depth analysis with strong**	**Good analysis with logical reasoning**	**Basic analysis with limited depth**	**Weak or no analysis; lacks reasoning**	

		reasoning and insights				
Solution Design	20	**Innovative, feasible solutions with strong justification**	**Logical solutions with adequate justification**	**Basic solutions with weak justification**	**Incorrect or no solution provided**	
Clarity	10	**Clear, well-structured, and easy to understand**	**Organized and understandable presentation**	**Limited clarity and structure**	**Poor clarity; difficult to understand**	

QUESTION 1

A food delivery company wants to know the average delivery time in Chennai. It randomly selects 100 deliveries and obtains an average delivery time of 32 minutes. What population parameter is being estimated, and what is the point estimate?

Solution:

Population: All deliveries made by the food delivery company in Chennai, and the delivery time associated with each of those deliveries.

Sample: The 100 randomly selected deliveries used to calculate the average delivery time.

Parameter of interest: The population mean delivery time, μ, measured in minutes, across all deliveries made by the company in Chennai.

Point estimate: The sample mean delivery time, $\bar{x}$, is the natural point estimate of the population mean μ.

Calculation:

Given: $n = 100$, $\bar{x} = 32$ minutes.

Since the sample mean is directly reported, the point estimate of μ is:

$\bar{x}$ = 32 minutes.

Interpretation: Based on the sample of 100 deliveries, the best single-valued estimate of the average delivery time for all deliveries made by the company in Chennai is 32 minutes. This value is only a point estimate; it does not, by itself, indicate how much the estimate might vary from sample to sample.

Scenario-based interpretation: For operational planning, 32 minutes can be used as a working benchmark for the company's average delivery performance in Chennai. However, a single point estimate does not reveal delivery-time variability across zones, time of day, or traffic conditions.

Practical recommendation: The company should accompany this point estimate with a margin of error or confidence interval, and should track the metric across a larger and more representative set of deliveries before using it to set service-level agreements or driver-allocation targets.

QUESTION 2

An e-commerce website wants to estimate the percentage of customers who purchase a product after viewing its advertisement. From 800 randomly selected visitors, 176 made a purchase. Determine the point estimate of the population proportion.

Solution:

Population: All visitors who view the advertisement on the e-commerce website, and whether or not each visitor goes on to make a purchase.

Sample: The 800 randomly selected visitors who viewed the advertisement, of whom 176 made a purchase.

Parameter of interest: The population proportion, p, of visitors who purchase a product after viewing the advertisement.

Point estimate: The sample proportion, $\hat{p}$, is the natural point estimate of the population proportion p.

Calculation:

Given: $n = 800$, $x = 176$ (number who made a purchase).

$\hat{p} = x / n = 176 / 800 = 0.22$.

Expressed as a percentage: $\hat{p} = 22\%$.

Interpretation: Based on this sample, the best single-valued estimate of the population purchase rate after viewing the advertisement is 22%. This is the observed rate in the sample and serves as the point estimate of the true, unknown population proportion.

Scenario-based interpretation: A 22% conversion rate suggests the advertisement is prompting roughly one in five viewers to purchase. As a point estimate, it does not indicate the precision of this figure or how it might vary if a different set of 800 visitors were sampled.

Practical recommendation: The website should construct a confidence interval around this proportion to understand the plausible range of the true conversion rate, and should monitor the rate across different visitor segments, devices, and time periods before drawing firm conclusions about advertisement performance.

QUESTION 3

A bank wants to estimate the average monthly expenditure of its credit-card customers. A random sample of 150 customers has a mean expenditure of ₹24,500. Explain how the sample mean can be used as a frequentist estimate of the population mean.

Solution:

Population: All credit-card customers of the bank, and their respective monthly expenditures.

Sample: The 150 randomly selected credit-card customers, whose expenditures produced a sample mean of ₹24,500.

Parameter of interest: The population mean monthly expenditure, μ, of all the bank's credit-card customers.

Point estimate: The sample mean, $\bar{x}$ = ₹24,500, is the point estimate of μ.

Frequentist approach:

In the frequentist framework, the population mean μ is treated as a fixed, unknown constant, not a random variable. Probability statements are instead made about the estimator itself, based on how it would behave under repeated, hypothetical sampling from the same population.

The sample mean $\bar{x}$ is a random variable: its value changes from sample to sample because it depends on which 150 customers happen to be selected. $\bar{x}$ is used as a frequentist estimator of μ because it has three properties that are evaluated over repeated sampling:

Unbiasedness: The expected value of $\bar{x}$ over repeated samples equals μ, i.e., $E(\bar{x}) = \mu$, so $\bar{x}$ does not systematically overestimate or underestimate the population mean.

Consistency: As the sample size n increases, $\bar{x}$ converges to μ, in line with the law of large numbers.

Efficiency: Under standard assumptions, $\bar{x}$ has the smallest variance among unbiased estimators of μ, making it a reliable single-value estimate.

Interpretation: The value ₹24,500 is the specific realization of the estimator $\bar{x}$ obtained from this particular sample of 150 customers. A frequentist does not interpret ₹24,500 as a probability statement about μ itself; instead, the estimate is trusted because, if the sampling process were repeated many times, the average of all such sample means would equal the true population mean μ.

Scenario-based interpretation: For the bank, ₹24,500 is a reasonable working estimate of average monthly credit-card expenditure across its customer base, but the frequentist justification for using it rests on the sampling procedure being repeatable and unbiased, not on any single sample being guaranteed correct.

Practical recommendation: The bank should report the point estimate together with its standard error or a confidence interval, so that the precision of the estimate — grounded in the frequentist idea of variability across repeated samples — is communicated alongside the estimate itself.

QUESTION 4

A streaming platform selects 500 subscribers to estimate the percentage of users who watch movies every day. If 320 users report watching movies daily, calculate the point estimate and explain its interpretation.

Solution:

Population: All subscribers of the streaming platform, and whether or not each subscriber watches movies every day.

Sample: The 500 randomly selected subscribers, of whom 320 reported watching movies daily.

Parameter of interest: The population proportion, p, of subscribers who watch movies every day.

Point estimate: The sample proportion, $\hat{p}$, is the natural point estimate of the population proportion p.

Calculation:

Given: $n = 500$, $x = 320$ (number of users watching movies daily).

$\hat{p} = x / n = 320 / 500 = 0.64$.

Expressed as a percentage: $\hat{p} = 64\%$.

Interpretation: Based on this sample, the best single-valued estimate of the proportion of all subscribers who watch movies daily is 64%. This figure is derived directly from the sample data and represents the platform's current best guess of the true population proportion.

Scenario-based interpretation: A 64% daily-viewing rate suggests strong day-to-day engagement among subscribers. As a point estimate, however, it gives no information about how precise this figure is or how it might change with a different sample of 500 subscribers.

Practical recommendation: The platform should pair this point estimate with a confidence interval to express the uncertainty around it, and should examine whether the daily-viewing rate differs across subscription tiers, devices, or content genres before using it to guide content or engagement strategy.

QUESTION 5

A manufacturing company randomly selects 64 smartphones and finds that their average battery life is 18.5 hours with a standard deviation of 2 hours. Construct a 95% confidence interval for the average battery life.

Solution:

Population: All smartphones manufactured by the company, and their respective battery lives.

Sample: The 64 randomly selected smartphones, with sample mean battery life 18.5 hours and sample standard deviation 2 hours.

Parameter of interest: The population mean battery life, μ, measured in hours.

Point estimate: The sample mean, $\bar{x} = 18.5$ hours, is the point estimate of μ.

Calculation:

Given: $n = 64$, $\bar{x} = 18.5$ hours, $s = 2$ hours, confidence level = 95%.

Because the sample size is reasonably large ($n = 64$), the standard normal (z) critical value is used: $z = 1.96$.

Standard error: $SE = s / \sqrt{n} = 2 / \sqrt{64} = 2 / 8 = 0.25$ hours.

Margin of error: $E = z \times SE = 1.96 \times 0.25 = 0.49$ hours (rounded to two decimal places).

Confidence interval: $\bar{x} \pm E = 18.5 \pm 0.49$.

Lower limit: $18.5 - 0.49 = 18.01$ hours.

Upper limit: $18.5 + 0.49 = 18.99$ hours.

95% confidence interval: (18.01 hours, 18.99 hours).

Confidence interval interpretation: We are 95% confident that the true population mean battery life of the company's smartphones lies between 18.01 hours and 18.99 hours. This means that if the sampling process were repeated many times and a 95% confidence interval were constructed each time, approximately 95% of those intervals would contain the true population mean.

Scenario-based interpretation: The relatively narrow interval (about one hour wide) indicates that the sample of 64 smartphones gives a fairly precise estimate of average battery life. The company can reasonably claim an average battery life close to 18.5 hours, with the true value very likely falling between 18.01 and 18.99 hours.

Practical recommendation: The company can use this interval to support marketing or specification claims about battery life, but should continue to test battery performance across different usage patterns, device batches, and ageing conditions, since the interval reflects only the variability captured in this particular sample.